Program :-1

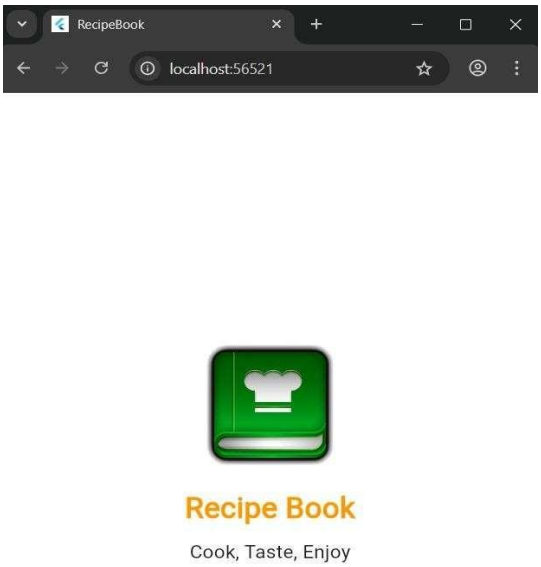
Defination : Install Flutter SDK and Android Studio, create a new Flutter project named recipe_book_app, and design a splash screen with text and app logo.

Code :

```
import 'package:flutter/material.dart';
import 'home_screen.dart';

class SplashScreen extends StatefulWidget {
  const SplashScreen({super.key});

  @override
  State<SplashScreen> createState() => _SplashScreenState();
}

class _SplashScreenState extends State<SplashScreen> {

  @override
  void initState() {
    super.initState();

    Future.delayed(const Duration(seconds: 3), () {
      Navigator.pushReplacement(
        context,
        MaterialPageRoute(
          builder: (context) => const HomeScreen(),
        ),
      );
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.white,
```

```
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [

          Image.asset(
            "assets/images/recipe-book.png",
            height: 120,
          ),

          const SizedBox(height: 20),

          const Text(
            "Recipe Book",
            style: TextStyle(
              fontSize: 28,
              fontWeight: FontWeight.bold,
              color: Colors.orange,
            ),
          ),

          const SizedBox(height: 10),

          const Text(
            "Cook, Taste, Enjoy",
            style: TextStyle(fontSize: 18),
          ),

        ],
      ),
    ),
  );
 }
}
```

Output :



Program :-2

Defination : Create a Home Screen containing three buttons: All Recipes, Add Recipe, and Categories.

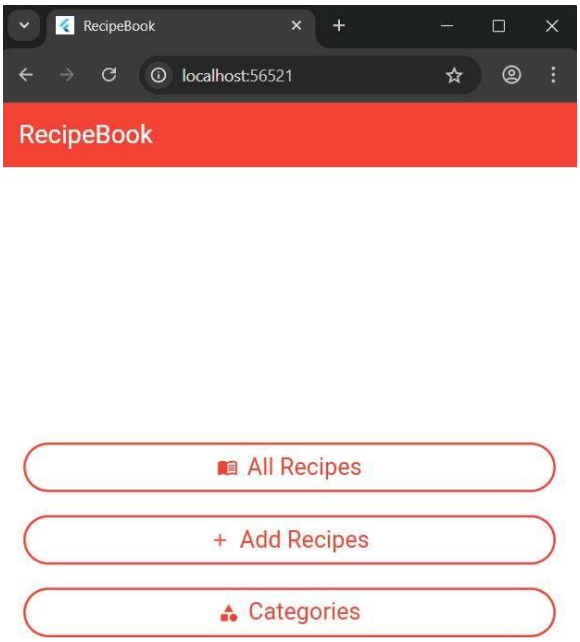
Code :

```
import 'package:flutter/material.dart';
import 'all_recipes_screen.dart';
import 'add_recipe_screen.dart';
import 'categories_screen.dart';

class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor:  Colors.white,

      appBar: AppBar(
        title: const Text("RecipeBook"),
        backgroundColor: Colors.red,
        foregroundColor: Colors.white,
      ),

      body: Padding(
        padding: const EdgeInsets.all(20.0),

        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          crossAxisAlignment: CrossAxisAlignment.stretch,
          children: [


            ElevatedButton.icon(
              icon: const Icon(Icons.menu_book, color: Colors.red),
              label: const Text(
                "All Recipes",
```

```
    style: TextStyle(fontSize: 20, color: Colors.red),
  ),

  style: ElevatedButton.styleFrom(
    padding: const EdgeInsets.symmetric(vertical: 15),
    backgroundColor: Colors.white,
    side: const BorderSide(color: Colors.red, width: 2),
    elevation: 0,
  ),

  onPressed: () {
    Navigator.push(
    context,
    MaterialPageRoute(
      builder: (context) => const AllRecipesScreen(),
    ),
    );
  },
),

const SizedBox(height: 20),


ElevatedButton.icon(
  icon: const Icon(Icons.add, color: Colors.red),
  label: const Text(
    "Add Recipes",
    style: TextStyle(fontSize: 20, color: Colors.red),
  ),

  style: ElevatedButton.styleFrom(
    padding: const EdgeInsets.symmetric(vertical: 15),
    backgroundColor: Colors.white,
    side: const BorderSide(color: Colors.red, width: 2),
    elevation: 0,
  ),

  onPressed: () {
    Navigator.push(
    context,
```

```
          MaterialPageRoute(
            builder: (context) => const AddRecipeScreen(),
          ),
        );
      },
    ),

    const SizedBox(height: 20),


    ElevatedButton.icon(
      icon: const Icon(Icons.category, color: Colors.red),
      label: const Text(
        "Categories",
        style: TextStyle(fontSize: 20, color: Colors.red),
      ),

      style: ElevatedButton.styleFrom(
        padding: const EdgeInsets.symmetric(vertical: 15),
        backgroundColor: Colors.white,
        side: const BorderSide(color: Colors.red, width: 2),
        elevation: 0,
      ),

      onPressed: () {
        Navigator.push(
        context,
        MaterialPageRoute(
            builder: (context) => const CategoriesScreen(),
          ),
        );
      },
    ),

  ],
 ),
),
);
}
}
```

Output :


RecipeBook
localhost:56521
RecipeBook
All Recipes
+ Add Recipes
Categories

Program :-3

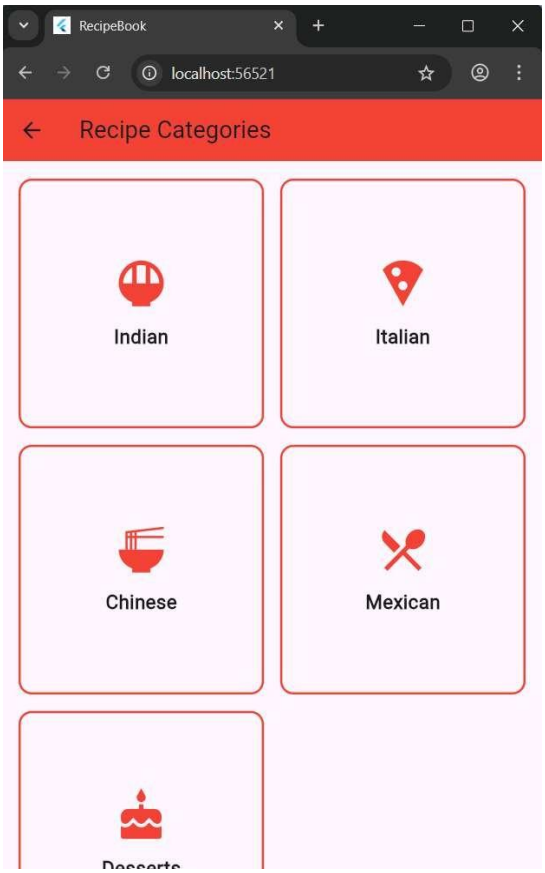
Defination : Display recipe categories such as Indian, Italian, Chinese, Mexican, and Desserts using icons in a grid layout.

Code :

```
import 'package:flutter/material.dart';

class CategoriesScreen extends StatelessWidget {
  const CategoriesScreen({super.key});

  @override
  Widget build(BuildContext context) {

    final List<Map<String, dynamic>> categories = [
      {"name": "Indian", "icon": Icons.rice_bowl},
      {"name": "Italian", "icon": Icons.local_pizza},
      {"name": "Chinese", "icon": Icons.ramen_dining},
      {"name": "Mexican", "icon": Icons.local_dining},
      {"name": "Desserts", "icon": Icons.cake},
    ];

    return Scaffold(
      appBar: AppBar(
        title: const Text("Recipe Categories"),
        backgroundColor: Colors.red,
      ),

      body: GridView.builder(
        padding: const EdgeInsets.all(16),
        itemCount: categories.length,

        gridDelegate: const SliverGridDelegateWithFixedCrossAxisCount(
          crossAxisCount: 2,
          crossAxisSpacing: 15,
          mainAxisSpacing: 15,
        ),
```

```
itemBuilder: (context, index) {
  return Container(
  decoration: BoxDecoration(
    border: Border.all(color: Colors.red, width: 2),
    borderRadius: BorderRadius.circular(12),
  ),

  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [

      Icon(
        categories[index]["icon"],
        size: 50,
        color: Colors.red,
      ),

      const SizedBox(height: 10),

      Text(
        categories[index]["name"],
        style: const TextStyle(
        fontSize: 18,
          fontWeight: FontWeight.bold,
        ),
      ),

    ],
  ),
 );
},
),
);
}
}
```

Output :


RecipeBook
localhost:56521
Recipe Categories
Indian
Italian
Chinese
Mexican

Program :-4

Defination : Build a form with fields for Recipe Name, Ingredients, Steps, Category, Time Required, and optional image upload, including proper form validation.

Code :

```
import 'package:flutter/material.dart';
import 'dart:io';
import 'package:image_picker/image_picker.dart';

class AddRecipeScreen extends StatefulWidget {
  const AddRecipeScreen({super.key});

  @override
  State<AddRecipeScreen> createState() => _AddRecipeScreenState();
}

class _AddRecipeScreenState extends State<AddRecipeScreen> {

  final _formKey = GlobalKey<FormState>();

  final TextEditingController nameController = TextEditingController();
  final TextEditingController ingredientsController = TextEditingController();
  final TextEditingController stepsController = TextEditingController();
  final TextEditingController timeController = TextEditingController();

  String? selectedCategory;
  File? selectedImage;

  final List<String> categories = [
    "Indian",
    "Italian",
    "Chinese",
    "Mexican",
    "Desserts"
  ];
```

```
Future<void> pickImage() async {
  final pickedFile =
  await ImagePicker().pickImage(source: ImageSource.gallery);

  if (pickedFile != null) {
    setState(() {
      selectedImage = File(pickedFile.path);
    });
  }
}

void saveRecipe() {
  if (_formKey.currentState!.validate()) {

    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(
        content: Text("Recipe Saved Successfully"),
      ),
    );

    Navigator.pop(context);
  }
}

@override
Widget build(BuildContext context) {

  return Scaffold(
    appBar: AppBar(
      title: const Text("Add Recipe"),
      backgroundColor: Colors.orange,
    ),

    body: SingleChildScrollView(
      padding: const EdgeInsets.all(16),

      child: Form(
        key: _formKey,

        child: Column(
```

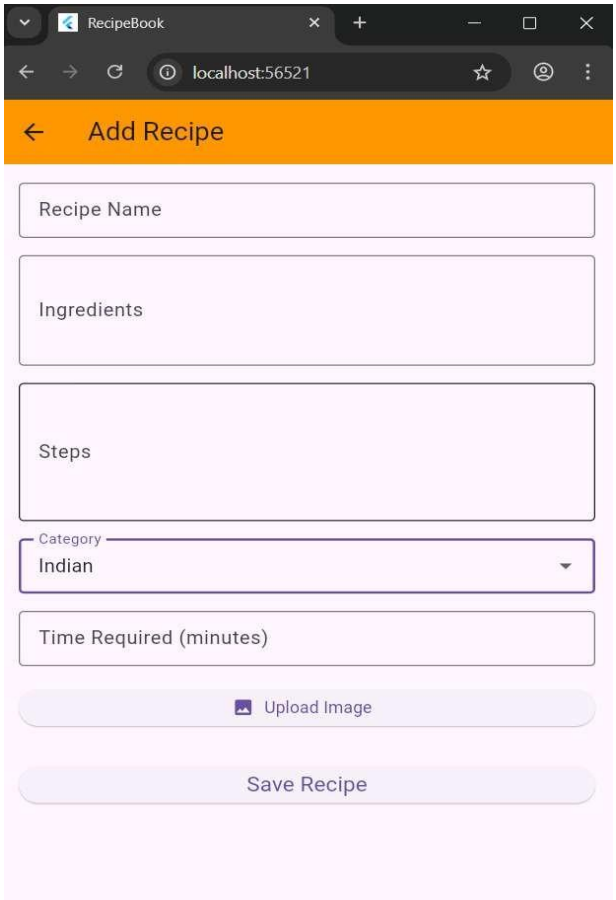
```
crossAxisAlignment: CrossAxisAlignment.stretch,
children: [


 TextFormField(
  controller: nameController,
  decoration: const InputDecoration(
  labelText: "Recipe Name",
    border: OutlineInputBorder(),
  ),
  validator: (value) =>
  value!.isEmpty ? "Enter recipe name" : null,
 ),

 const SizedBox(height: 15),


 TextFormField(
  controller: ingredientsController,
  maxLines: 3,
  decoration: const InputDecoration(
    labelText: "Ingredients",
    border: OutlineInputBorder(),
  ),
  validator: (value) =>
  value!.isEmpty ? "Enter ingredients" : null,
 ),

 const SizedBox(height: 15),


 TextFormField(
  controller: stepsController,
  maxLines: 4,
  decoration: const InputDecoration(
    labelText: "Steps",
    border: OutlineInputBorder(),
  ),
  validator: (value) =>
  value!.isEmpty ? "Enter steps" : null,
```

```
),

const SizedBox(height: 15),

DropdownButtonFormField<String>(
value: selectedCategory,
 decoration: const InputDecoration(
  labelText: "Category",
  border: OutlineInputBorder(),
 ),

 items: categories
    .map(
      (cat) => DropdownMenuItem(
    value: cat,
    child: Text(cat),
  ),
 )
    .toList(),

 onChanged: (value) {
  setState(() {
  selectedCategory = value;
  });
 },

 validator: (value) =>
 value == null ? "Select category" : null,
),

const SizedBox(height: 15),


TextFormField(
 controller: timeController,
 keyboardType: TextInputType.number,
 decoration: const InputDecoration(
 labelText: "Time Required (minutes)",
 border: OutlineInputBorder(),
```

```
          ),
          validator: (value) =>
          value!.isEmpty ? "Enter time required" : null,
        ),

        const SizedBox(height: 20),


        ElevatedButton.icon(
          icon: const Icon(Icons.image),
          label: const Text("Upload Image"),
          onPressed: pickImage,
        ),

        const SizedBox(height: 15),


        if (selectedImage != null)
          Image.file(
          selectedImage!,
            height: 150,
          ),

        const SizedBox(height: 20),

        ElevatedButton(
        onPressed: saveRecipe,
        child: const Text(
            "Save Recipe",
            style: TextStyle(fontSize: 18),
          ),
        ),

      ],
    ),
  ),
),
);
```

```
  }
}
```

Output :


RecipeBook
localhost:56521
Add Recipe
Recipe Name
Ingredients
Steps
Category
Indian
Time Required (minutes)
Upload Image
Save Recipe

Program :-5

Defination : Implement named routes to navigate between Home, Add Recipe, and Recipe List screens.

CODE -:

```
class HomeScreen extends StatelessWidget {
  const HomeScreen({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text("Recipe Book"),
        backgroundColor: Colors.orange,
        foregroundColor: Colors.white,
      ),
      body: Center(
        child: Padding(
          padding: const EdgeInsets.all(20),
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            crossAxisAlignment: CrossAxisAlignment.stretch,
            children: [
              ElevatedButton.icon(
```

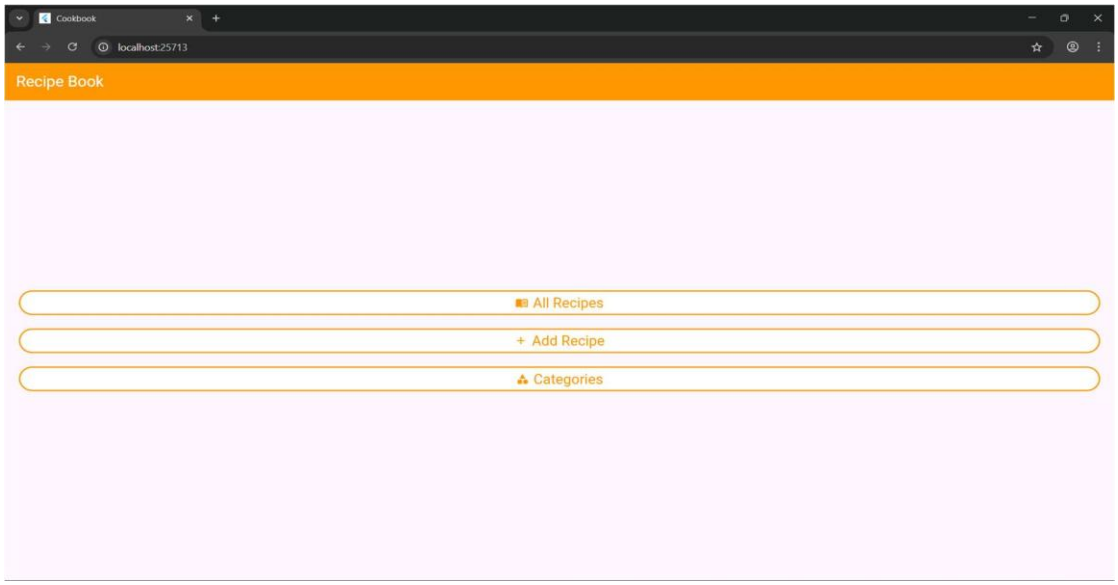
```
  icon: const Icon(Icons.menu_book, color: Colors.orange),
  label: const Text(
    "All Recipes",
    style: TextStyle(color: Colors.orange),
  ),
  style: ElevatedButton.styleFrom(
    padding: const EdgeInsets.symmetric(vertical: 15),
    textStyle: const TextStyle(fontSize: 20),
    backgroundColor: Colors.white,
    side: const BorderSide(color: Colors.orange, width: 2),
    elevation: 0,
  ),
  onPressed: () {
    Navigator.push(
    context,
    MaterialPageRoute(
      builder: (context) => const AllRecipeScreen(),
    ),
    );
  },
),
const SizedBox(height: 20),
ElevatedButton.icon(
  icon: const Icon(Icons.add, color: Colors.orange),
```

```
  label: const Text(
    "Add Recipe",
    style: TextStyle(color: Colors.orange),
  ),
  style: ElevatedButton.styleFrom(
    padding: const EdgeInsets.symmetric(vertical: 15),
    textStyle: const TextStyle(fontSize: 20),
    backgroundColor: Colors.white,
    side: const BorderSide(color: Colors.orange, width: 2),
    elevation: 0,
  ),
  onPressed: () {
    Navigator.push(
    context,
    MaterialPageRoute(
        builder: (context) => const AddRecipeScreen(),
      ),
    );
  },
),
const SizedBox(height: 20),
ElevatedButton.icon(
  icon: const Icon(Icons.category, color: Colors.orange),
  label: const Text(
```

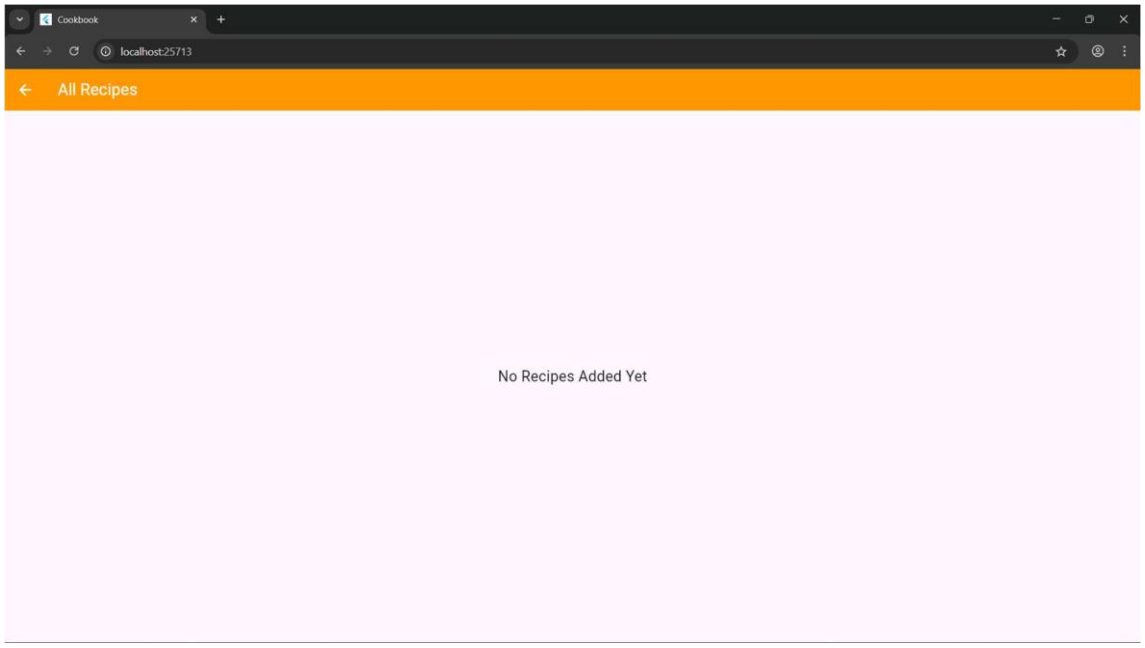
```
            "Categories",
            style: TextStyle(color: Colors.orange),
          ),
          style: ElevatedButton.styleFrom(
            padding: const EdgeInsets.symmetric(vertical: 15),
            textStyle: const TextStyle(fontSize: 20),
            backgroundColor: Colors.white,
            side: const BorderSide(color: Colors.orange, width: 2),
            elevation: 0,
          ),
          onPressed: () {
            Navigator.push(
            context,
            MaterialPageRoute(
              builder: (context) => const CategoriesScreen(),
            ),
            );
          },
        ),
      ],
    ),
  ),
),
);
```

```
  }

}
```

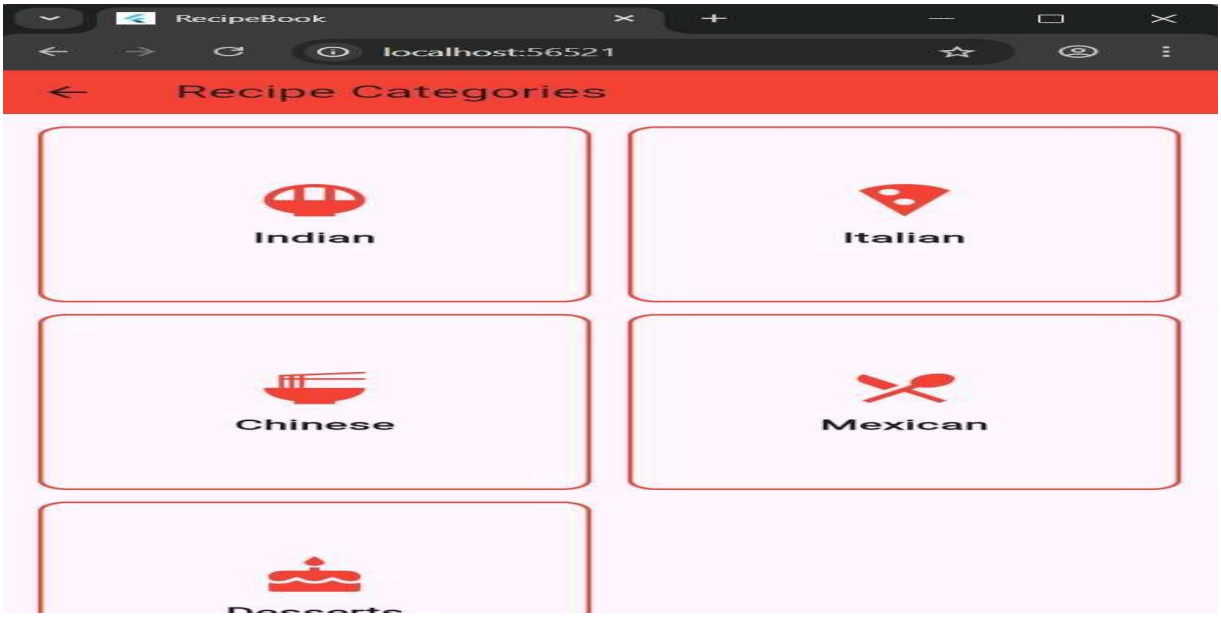
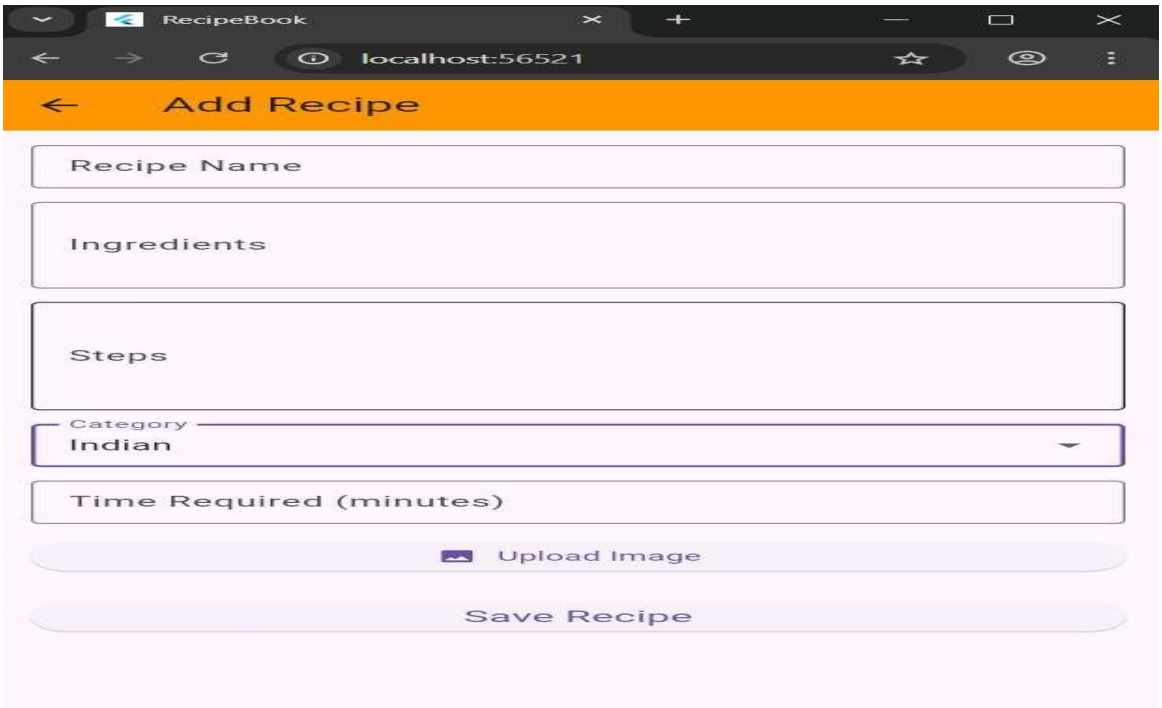
Output -:



After clicking all recipes



After clicking categories



After clicking add recipe